



V-SPARC: Detecting LPTs with VLITE

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Abstract

We conducted an extensive blind survey for new Long-Period Transients (LPTs), searching VLITE data using automated filtering, manual image inspection, and catalog crossmatching. Though no new candidates were confirmed, the wide scope of our survey established valuable baseline constraints for future transient searches and anchored an observational sky rate of LPT bursts in the galactic plane.

Introduction

Long-Period Radio Transients (LPTs)

- Variable on periods of minutes to hours
- Bright, often-polarized coherent radio bursts
- Fewer than 30 confirmed sources

Importance

- Unknown origin; current research suggests binary systems or magnetars
- Potential for deeper understanding of radio transient groups

Why VLITE?

- Wide-field radio monitoring system
- Sensitive to minute-to-hour timescale variability
- Includes known LPTs in observation data



Figure 1. The Karl Jansky Very Large Array (VLA) in New Mexico, on which the auxiliary telescopes of VLITE are located

Research Objectives

- Use VLITE A, B, and C configurations to look for LPTs
- Investigate other serendipitous discoveries
- Refine techniques for filtering images
- Determine constraints on future LPT searches

Methodology

Data and Tools

- Catalogs from the A, B, and C configurations (configs) of VLITE
- PyBDSF, DS9, TOPCAT, Aladin, and Jupyter

Filtering LPT Candidates

- Filtered for a signal-to-noise ratio (S/N) > 10, compactness > 0.9, distance from center < image radius (1.25°, 1.75°, and 3° for A, B, and C configs)
- Required separation from known sources in a variety of radio catalogs
- For C config, required 80" separation from the nearest in-catalog neighbor to eliminate sidelobes
- Images containing >1 transient candidate were cut (likely artifact-dominated)

Image Analysis

- Visibility functions of known sources were subtracted to remove associated artifacts
- Visual inspection of FITS files was used to determine veracity of detections
- Promising sources were investigated for variability over time using PyBDSF analysis on TGSS/NVSS survey data

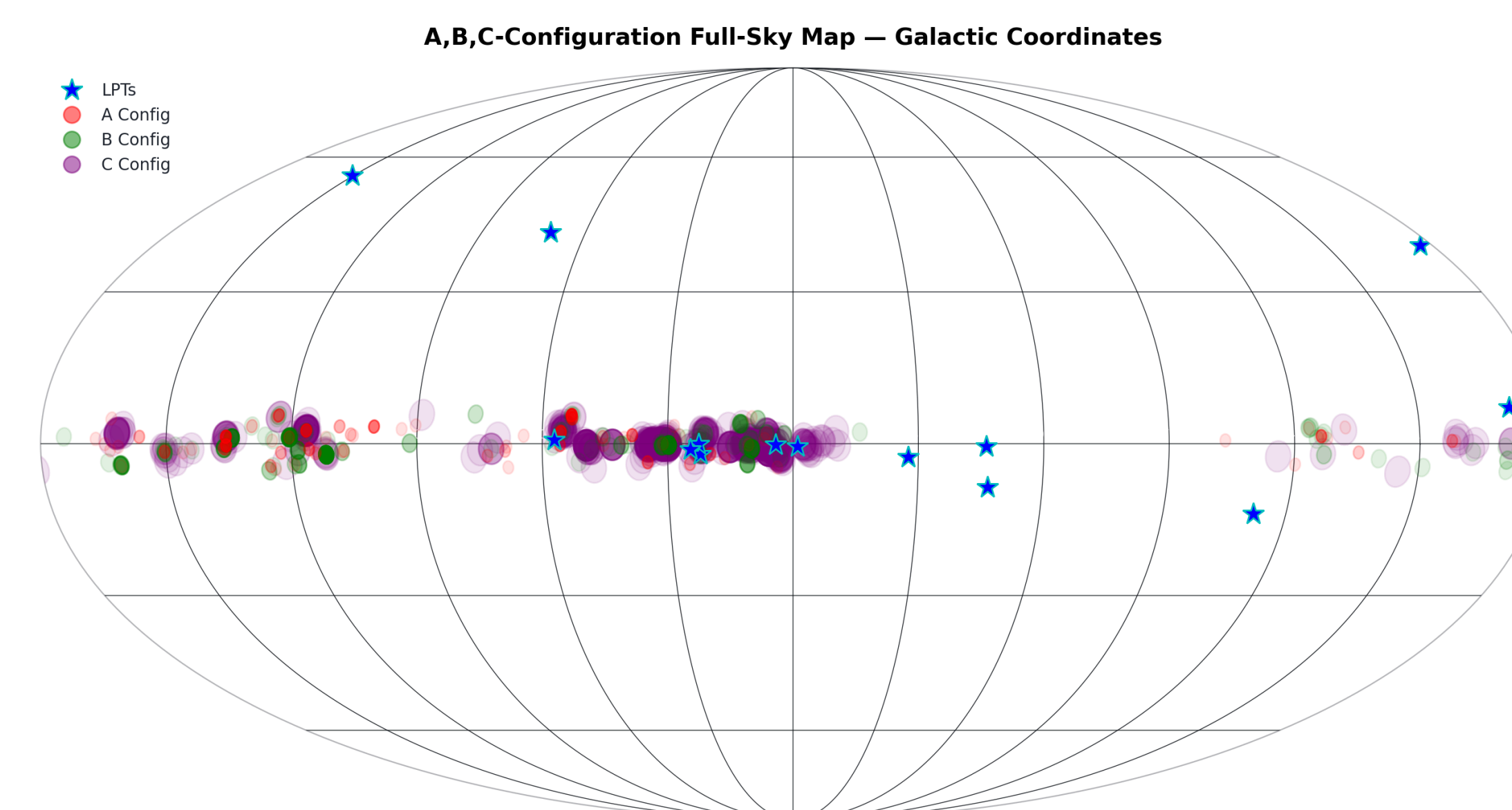


Figure 2. Galactic coordinate map of the A, B, and C Config candidates (206, 282, and 647 respectively) and known LPTs

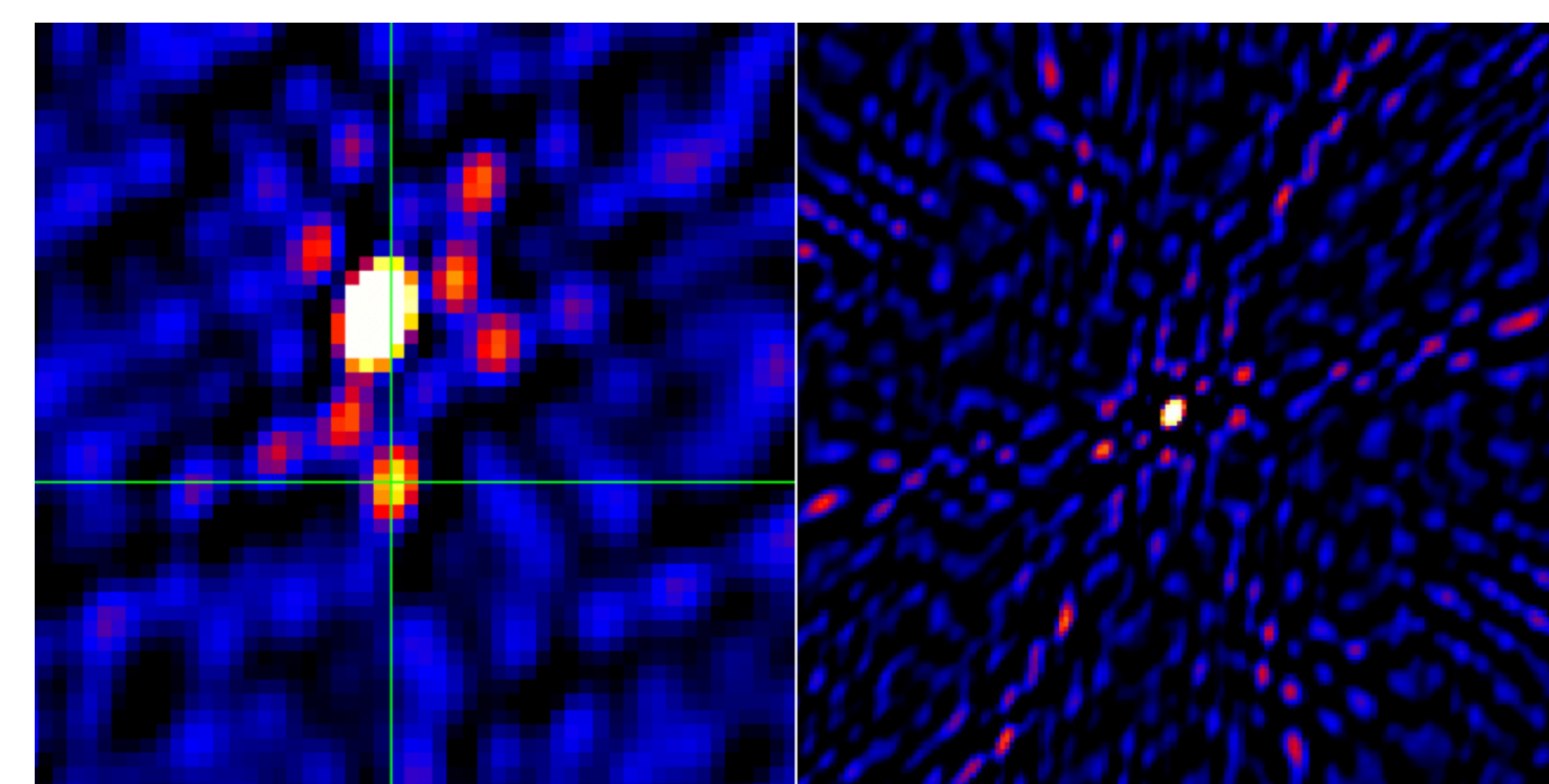


Figure 3. A sidelobe (marked by crosshair) of a bright source

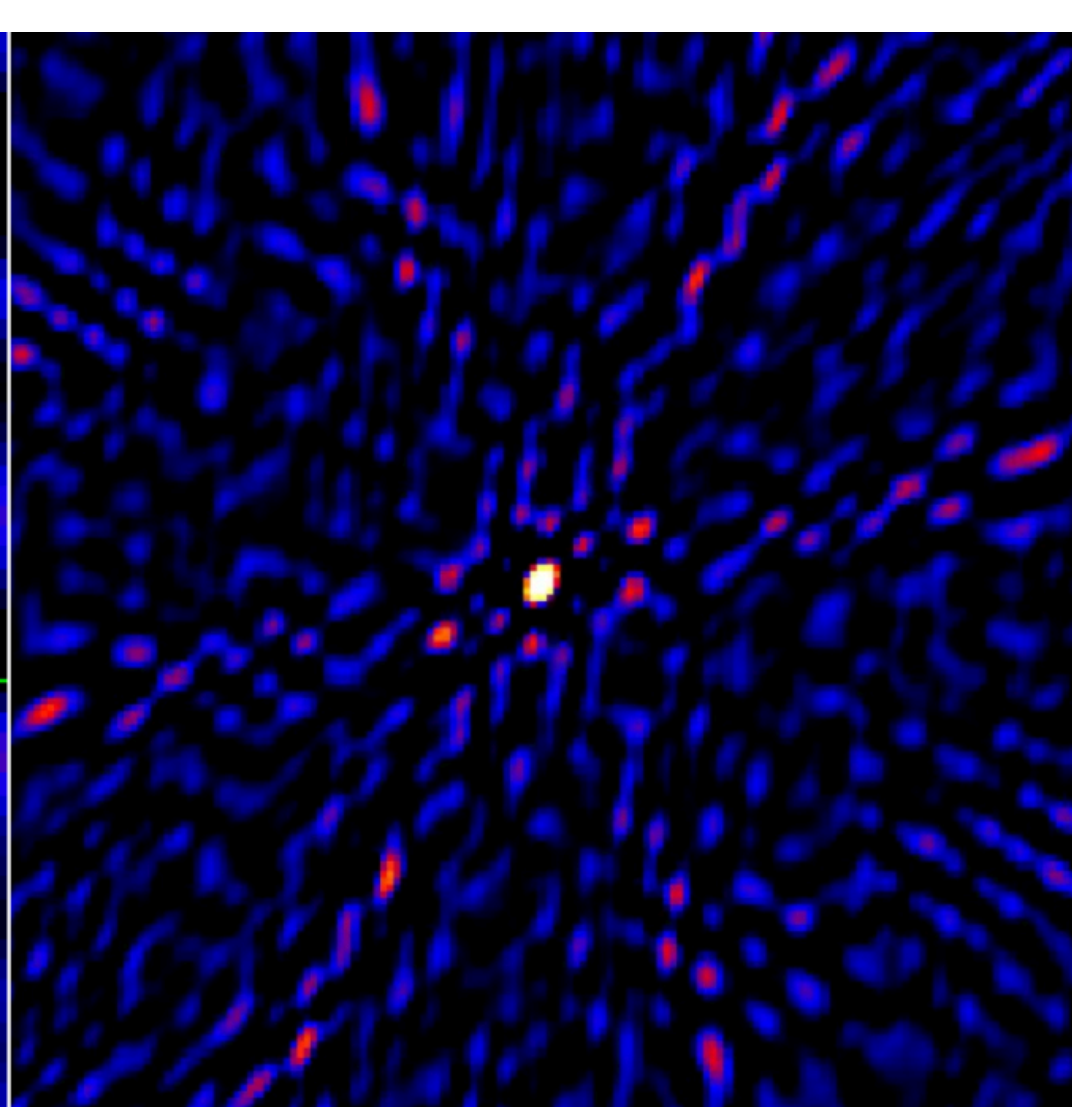


Figure 4. Confirmed LPT detection (source at center)

Results

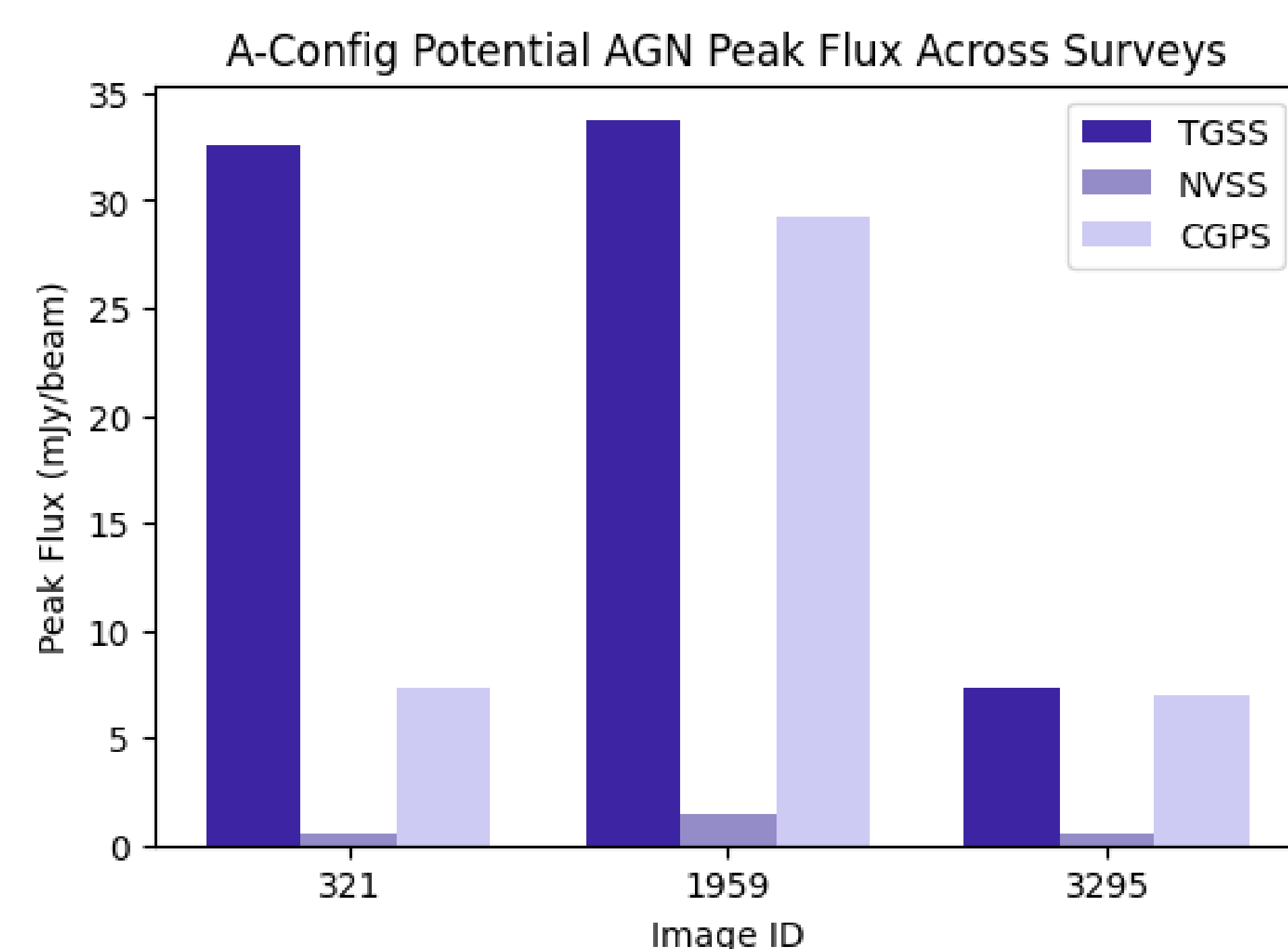


Figure 5. Sources demonstrating variable peak flux consistent with the behavior of recently activated AGNs.

- Calculated limiting (minimum detectable) fluences across all good images and estimated areal LPT burst density at the median limiting fluences of each configuration from three C config detections
- True burst density likely $\sim 1 \times 10^{-5}$ per deg^2 , implying $\sim 100,000 \text{ deg}^2$ to detect a bright burst

- Survey covered $250,000 \text{ deg}^2$ across 20,000 images and 1293 hr of sky time
- Highly artifact-limited; frequent sidelobes and ambiguous sources in noisy sky regions
- Three of four known LPTs in data passed filtering: ASKAP J183950.5-075635.0 and ASKAP/DART J1832-0911 (C config), GPM J1839-10 (B config)
- Three serendipitous C config detections, none in A/B
- Three A config sources consistent with variable to recently activated active galactic nucleus (AGN) systems

Discussion

- Limiting fluences and areal density of bursts help fix how much coverage is needed to detect larger numbers of LPTs

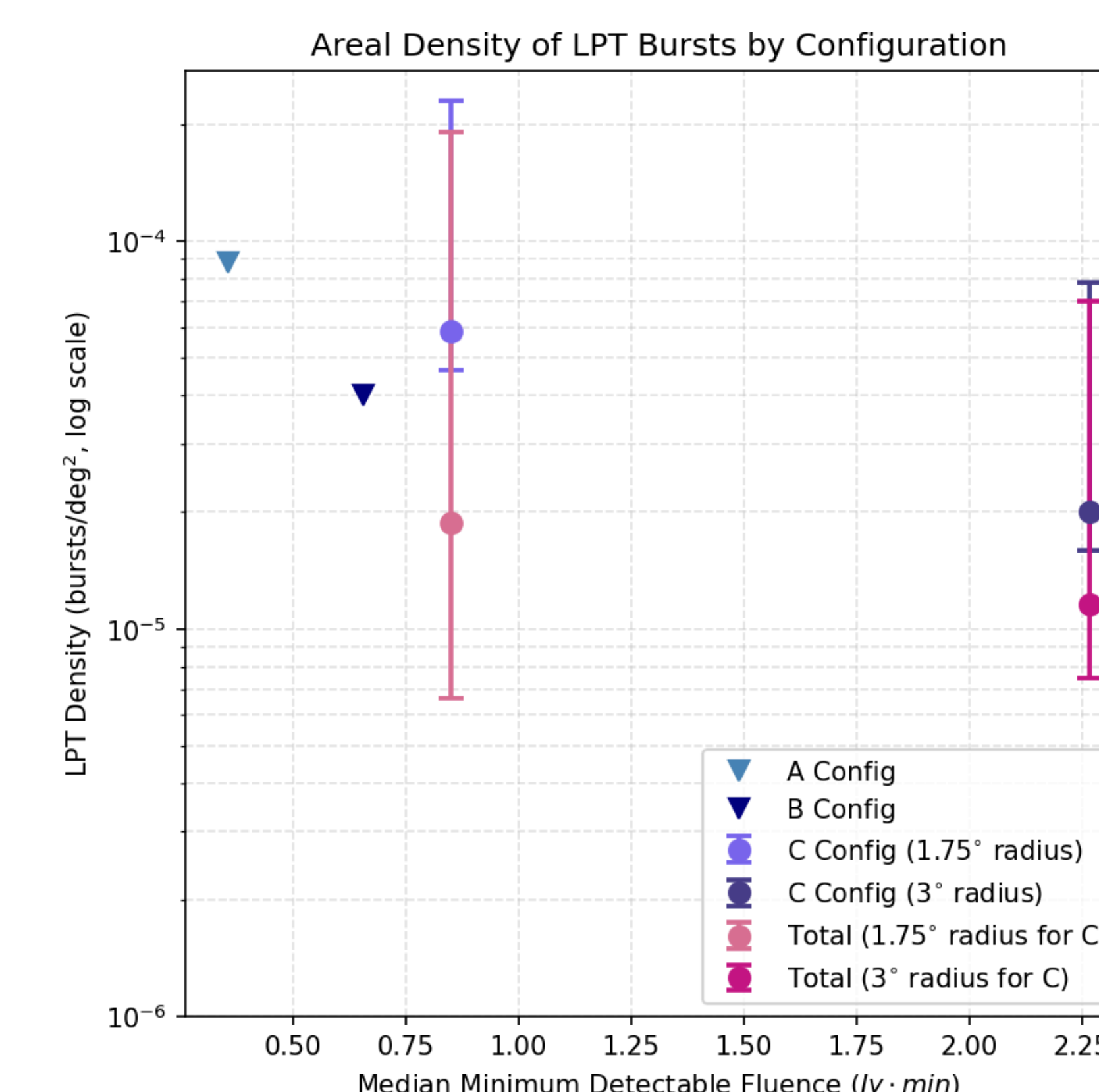
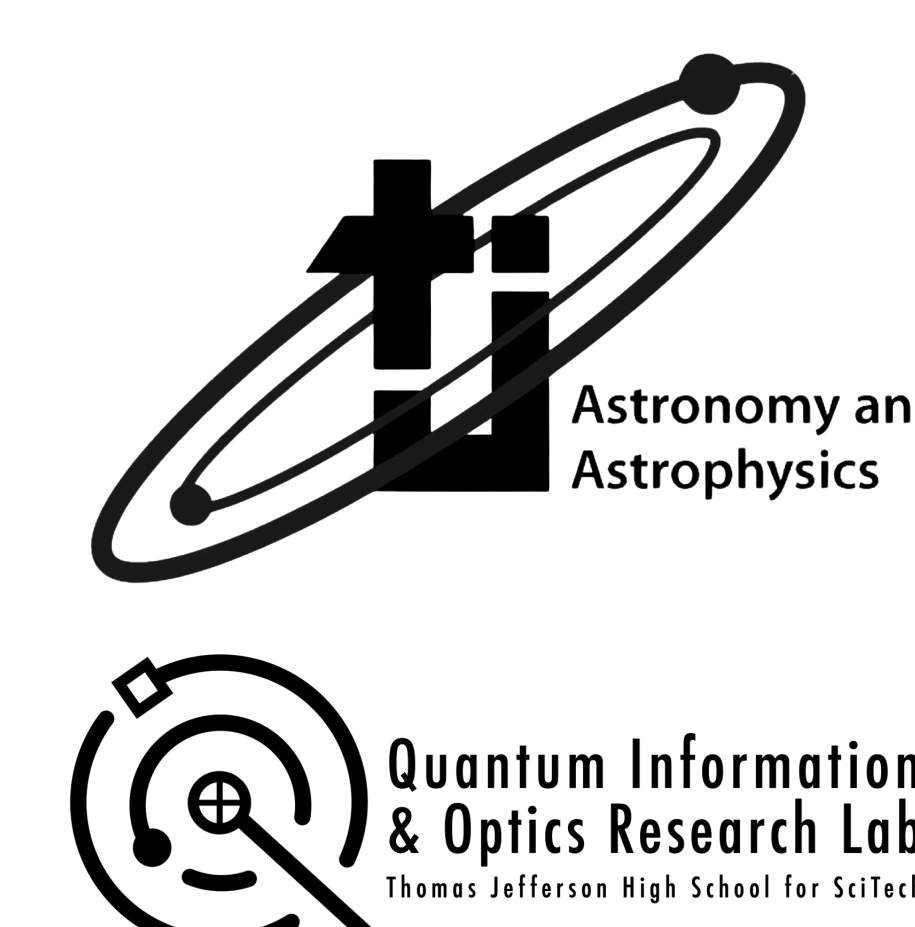


Figure 6. Areal burst densities with error from Poisson analysis at limiting fluence of each config

Future Work

- Use limiting fluences and areal density constraints to guide the extent and precision of future surveys
- Translate the pipeline structure to ongoing LPT surveys
- Investigate properties of turned-on AGNs as discovered by such searches
- Examine potential variance in burst rate across galactic latitude/longitude

References



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